Performance Engineering of a PySpark Pipeline on NYC Taxi Trips

Gabriel Matias Falcão (77390)
NOVA School of Science and Technology, Universidade NOVA de Lisboa
Lisbon, Portugal

Abstract

We profile and optimise a PySpark pipeline over 29.7 M NYC yellow-taxi trips (three mandatory operations plus origin-destination flows and trip-type classification). Following a profile, hypothesise, optimise, measure loop, the baseline's cost is traced to sort-merge joins against tiny tables, spills in under-partitioned reduce stages and repeated scans. Broadcast joins, writing results once, materialising the cleaned data and explicitly partitioning a cached result cut individual operations by up to 5×; caching the wrong DataFrames and smaller scan splits are reported as measured failures. Execution time scales sub-linearly with input size because of a ∼25 s fixed cost, and a single-machine Docker cluster shows why adding executors on shared hardware does not help.

Keywords

Apache Spark, performance engineering, query plans, broadcast join, caching

1 Introduction

The pipeline processes the NYC Taxi and Limousine Commission (TLC) yellow taxi trip records [3]: one parquet file per month plus a 265-row zone lookup table. Development and all measurements use 2026-01..08 (29.7 M records, 488 MB of parquet). Besides the mandatory operations (M1 cleaning statistics, M2 temporal aggregation per borough, day and hour, M3 monthly zone statistics with rolling windows), two optional ones were chosen for the bottlenecks they expose. **O2** (origin-destination flows) joins the zone table twice and derives three outputs (route shares, top-20 per borough, Q1/Q3 reversals) from one monthly aggregate, so it stresses multi-way joins and recomputation. **O4** (trip-type classification) aggregates trips into at most 263×24 buckets and joins that small result back to every trip, the shape where join strategy and hot-zone skew decide the cost.

Every operation joins ∼21 M trips with a tiny table and aggregates, so the cost is how Spark moves and re-reads data, not arithmetic. Code, logs and per-optimisation reports are in the repository [2].

2 Pipeline

All operations share two steps. *Load* groups the files by physical parquet schema (2026 has two variants), casts each group to the documented schema with `try_cast` and combines them with `unionByName`. *Clean* keeps a record only if it passes the M1 rules: no null in a used column, fare and distance > 0, 1–6 passengers, dropoff after pickup, location ids in 1–263 (20.95 M records remain). Each operation then joins the zone table and aggregates; M3 and O2 add windows, O2 a pivot, O4 a second aggregation after the join back.

Op	Wall (s)	Stages	Tasks	Shuffle read	Spill mem / disk
M1	14.5	8	76	5.4 KB	0
M2	45.7	18	152	953.8 MB	1.9 GB / 832.6 MB
M3	21.1	9	76	318.5 MB	768 MB / 261 MB
O2	20.3	81	515	53.9 MB	0
O4	22.1	18	220	222.4 MB	704 MB / 107 MB

Table 1: Baseline, median of 3 runs (stage, task and shuffle counts identical in all runs).

Design decisions: the DataFrame API is used throughout, because operations are composed in Python from shared pieces (cleaning rules, zone joins, feature flags), and it compiles to the same Catalyst plans as SQL; no UDFs are used, so no work goes through Python workers; timestamps are read as `timestamp_ntz` with the session zone fixed to UTC, so hours and weekdays do not depend on the host; routes and zones are keyed by id because zone names are not unique. In the baseline `spark.sql.autoBroadcastJoinThreshold=-1`, so every join starts as a Sort-Merge Join (SMJ) and each optimisation is an explicit, measured feature flag. Figure 1 shows the physical plan of M2.

3 Profiling and Baseline

Setup. Laptop with an Intel i7-1195G7 (4 cores / 8 threads), 15 GiB RAM; Spark 4.2.0 (PySpark), OpenJDK 25, Python 3.14, `local[*]` (8 cores), `spark.driver.memory=2g`; non-default configuration: no automatic broadcast, session time zone UTC. Adaptive Query Execution (AQE) [7] is on with the default 200 shuffle partitions.

Method. Each configuration runs 3 times; reported times are medians. Outputs go to Spark's `noop` sink, because `show()` computes only the rows it prints. Every run logs `explain("formatted")` and `explain(True)`, and the Lab 3 profiler records jobs, stages, tasks, shuffle and spill per operation. Per-stage task-time quantiles come from the Spark UI REST API[5]. Spark's input metric reports 6 MiB for scans that read 383 MiB of parquet column chunks, so input volume is taken from the parquet footers instead.

Bottlenecks. (1) *Trip-side shuffle of the SMJ*: every operation hashes and sorts all ∼21 M cleaned trips to join a table of ≤ 6 312 rows (Figure 1, left). The stage after that shuffle is the most expensive of each operation and holds all of the spill (M2 stage 14: 476 MB read, 896 MB / 392 MB spilled, 99 s of task time). (2) *Too few reduce partitions*: AQE coalesces the 200 shuffle partitions into 9, so each task sorts ∼53 MB and spills, and 9 tasks run on 8 cores in two waves, the 9th alone for ∼12 s. (3) *Recomputation*: M2's display and parquet write are two actions, so its scan and join run twice (stages 11/14 rerun as 20/23, half of its shuffle); O2 rescans the trips for each of its 3 outputs and O4 scans them twice. (4) *Skew*: not found, the slowest reduce task is at most 1.1× the median, because

```
HashAggregate (final)
+- Exchange (Borough, day, hour)
   +- HashAggregate (partial)
      +- SortMergeJoin PULocationID=LocationID
         :- Sort
         :  +- Exchange (PULocationID) <- all trips
         :     +- Union (2 schema groups)
         :        :- Project(try_cast) / Filter
         :        :  +- Scan parquet [7 cols]
         :        +- Project(try_cast) / Filter
         :           +- Scan parquet [7 cols]
         +- Sort
            +- Exchange (LocationID)
               +- Scan csv (zones)
```

```
HashAggregate (final)
+- Exchange (Borough, day, hour)
   +- HashAggregate (partial)
      +- BroadcastHashJoin BuildRight
         :- Union (2 schema groups)
         :  :- Project(try_cast) / Filter
         :  :  +- Scan parquet [7 cols]
         :  +- Project(try_cast) / Filter
         :     +- Scan parquet [7 cols]
         +- BroadcastExchange
            +- Scan csv (zones)
```

Figure 1: M2 physical plan (`explain("formatted")`, simplified). Left: baseline, every cleaned trip is shuffled and sorted to join 265 rows. Right: after broadcasting the zone table (Section 4.1).

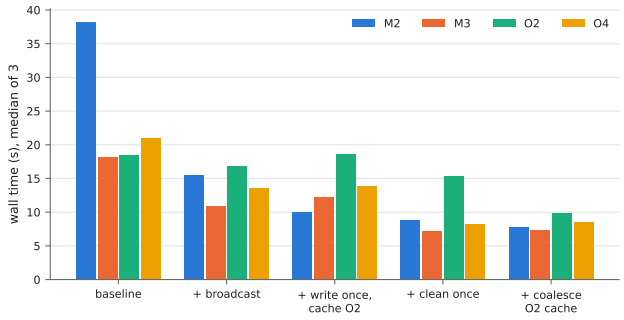


Figure 2: Wall time per operation across the cumulative optimisations, one session, median of 3 runs of all operations.

AQE packs many zones into each partition. (5) *Column pruning*: already done by Catalyst, which pushes the projection into the scan (7 of 20 columns read); selecting the used columns explicitly (`--prune-column`) leaves the `ReadSchema` unchanged. Three of the seven columns are read only for the cleaning rules, which Section 4.3 removes.

4 Optimisation Study

Optimisations are cumulative: each one is measured against the configuration that includes the previous ones, with all other settings unchanged. Table 2 summarises them, and Figure 2 re-measures the whole path in a single session (3 runs per configuration, each started on an idle machine), so the steps are directly comparable. It also shows that caching O2 alone did not pay off inside a full run (16.9 s → 18.6 s); only the coalesced cache did (9.8 s).

4.1 Broadcast the small side of every join

Bottleneck: the trip-side `Exchange` and sort of the SMJ and the spill after it (Section 3). *Hypothesis:* the zone table (265 rows) and O4's bucket table (≤ 6 312 rows) fit in every executor, so a Broadcast Hash Join (BHJ) can probe them inside the scan tasks and the trips never need to be shuffled. *Change:* `--broadcast-join` wraps the small side in a `broadcast()` hint [7]. *Result:* every `SortMergeJoin` becomes a `BroadcastHashJoin` (Figure 1, right), the 9-task reduce stage disappears, and only partial aggregates are shuffled. Shuffle read falls from 953.8 MB to 1.5 MB for M2 (318.5 MB to 0.7 MB for M3, 222.4 MB to 1.8 MB for O4), all spill goes to zero, and M2, M3 and O4 run 2.2×, 1.8× and 1.4× faster. O2 does not change, because its joins run on aggregated routes rather than trips.

4.2 Reusing results: cache versus write once

Bottleneck: with the shuffles gone, the scans dominate, and multi-action operations scan the trips once per action (M2 twice, O2 three times, O4 twice). *Hypothesis:* persisting the shared DataFrame leaves one scan per operation. *Change:* `--cache-dataframe` caches M2's result, O2's monthly routes and O4's cleaned trips. *Result:* only O2 improves (21.7 s → 17.0 s, 1.28×): its cache is a 4.8 MiB aggregate, so two of three scans disappear cheaply. M2 gets slower (20.9 s → 23.2 s): the cached result keeps all 200 shuffle partitions, which AQE would otherwise coalesce into one, so the write runs 200 tasks and produces 663 files instead of 6. O4 also gets slower (15.6 s → 19.9 s): building a 390 MiB in-memory cache of 21 M trips adds 41 s of task time to the scan, more than the second parquet scan it replaces. The cache was therefore kept for O2 only. For M2 the fix is to **write once** (`--write-once`): write the parquet output first and display the 967-row result read back from it. One of the two 18-task scan stages disappears, task time drops by 37 % and M2 goes from 23.0 s to 15.3 s (1.5×), with identical output.

4.3 Clean once, read many

Bottleneck: every operation still starts with a `Union` of two scans, each with `try_cast` projections and a 13-predicate cleaning filter; a run of all operations does this 9 times on the same rows and also reads columns used only by the cleaning rules. *Hypothesis:* materialising the cleaned, typed trips once as parquet removes that work from every operation, at the cost of one write; in-memory caching was ruled out by the O4 result and by the 2016–2020 data size. *Change:* `--materialise-clean` writes the cleaned trips (9 columns, 8 MiB row groups, 324 MiB) and runs M2–O4 on that copy; M1 still reads the raw files. *Result:* `Union` nodes drop from 9 to 1 and parquet scans from 18 to 10; M2's scan reads 5 columns with one pushed filter instead of 7 with 13. Task time falls by 41 % (M2), 60 % (M3), 19 % (O2) and 48 % (O4). The one-time write costs 19.4 s, yet a run of all operations already drops from 96.2 s to 92.3 s including it, and to 72.5 s once the copy exists.

4.4 Partition the O2 cache explicitly

Bottleneck: O2 is now the slowest operation, and its time is mostly fixed: 10.6 s on one month of data, 15.8 s on eight (Section 5). Its three outputs read the 4.8 MiB route cache in five 200-task stages whose

Optimisation	Bottleneck addressed	Measured impact (median wall time)
Broadcast joins	Trip shuffle + sort of the SMJ, spill	M2 45.7→20.9 s, M3 21.1→11.8 s, O4 22.1→15.6 s; spill 1.9 GB→0
Cache shared results	Trips scanned once per action	O2 21.7→17.0 s; M2 and O4 slower, kept for O2 only
Smaller scan splits	Uneven scan tasks	No gain, parallelism capped by 32 row groups
Write once (M2)	Result computed twice	M2 23.0→15.3 s, one scan stage less
Clean once, read many	Re-scan, cast and clean per operation	M2 13.8→9.0 s, M3 13.4→6.8 s, O2 19.0→16.3 s, O4 13.9→7.9 s; all ops 96.2→92.3 s incl. write
Coalesce O2 cache	200 tiny tasks per cache read	O2 24.2→14.8 s (run alone), tasks 2 443→370
All applied (one session)		M2 38.3→7.7 s (5.0×), M3 18.2→7.4 s (2.5×), O2 18.4→9.8 s (1.9×), O4 21.0→8.6 s (2.4×); all ops 136.8→76.9 s incl. clean write

Table 2: Summary of the optimisations; each row is measured against the configuration with the rows above it applied, in its own report. The last row comes from one session that ran every cumulative configuration 3 times (Figure 2).

tasks take 12–132 ms (median); these stages hold 74 s of O2's 128 s of task time. *Hypothesis:* the cache freezes the 200 shuffle partitions of the window that produces the routes, and AQE coalesces partitions when reading a shuffle but not when reading a cache, so each output pays task overhead for ~25 KB partitions (the same mechanism that slowed M2's cache in Section 4.2). *Change:* `--coalesce-cache` coalesces the routes to the default parallelism (8) before caching, without a shuffle. *Result:* tasks drop from 2 443 to 370, the five stages run 8 tasks of 0.3–1.7 s with 39 s instead of 74 s of task time, and O2 goes from 24.2 s to 14.8 s (1.64×, O2 run alone) with identical output. Smaller scan splits (`maxPartitionBytes`) gave no gain: the raw files hold only 32 row groups, which splits cannot divide [4, 6].

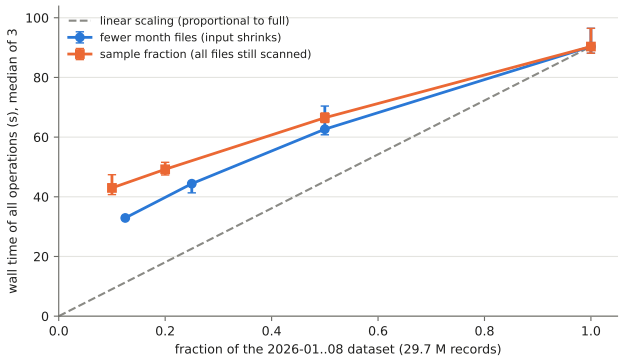


Figure 3: Wall time of all operations versus dataset fraction (median of 3, bars: min–max), optimised configuration, `local[*]`.

5 Analysis

Dataset scale. The optimised pipeline was run on 10, 20 and 50 % samples (the method suggested by the assignment) and on 1, 2, 4 and 8 month files (Figure 3). Time grows sub-linearly: 8× more input costs only 2.7× more time (32.9 s → 90.3 s). A straight line through the month series, $T \approx 25\,\text{s} + 66\,\text{s} \cdot f$, shows that about 25 s of every run is fixed: scheduling the ~10 actions per run and, above all, O2's ~2 300 small tasks over its cached routes (O2 takes 10.6 s on one month and 15.8 s on eight), since reduced by Section 4.4. The data-dependent part is linear, because after the optimisations no trip rows are shuffled and nothing spills at any size. Sampling understates small inputs: `sample()` drops rows after the scan, so a 10 % sample (43.0 s) is slower than a real 12.5 % input (32.9 s).

Cluster size. No multi-node cluster was available, so cluster size was studied on a Docker standalone cluster [8] on the same laptop (one master, workers of 2 cores / 2 GiB, driver in its own container), with the optimised pipeline on the full data (Table 3). Speedup stays far from linear and turns negative at 3 workers. The extra cores are not extra hardware: the same work costs more CPU as executors are added (M3 task time 8 s with one worker, 21 s with three), because

6 executor cores, driver and master compete for 4 physical cores, the clock drops with more active cores, and every executor Java Virtual Machine (JVM) repeats class loading and JIT compilation in a short run. The fixed driver-side part found above also caps the speedup (Amdahl [1]). On separate nodes the contention would disappear, so the scan-dominated part should scale much closer to linearly, while the fixed part would not.

Workers	Cores	Wall (s), median [min–max]	Speedup	Efficiency
1	2	69.3 [66.0–70.5]	1.00	1.00
2	4	64.2 [62.2–66.0]	1.08	0.54
3	6	74.0 [73.7–74.5]	0.94	0.31

Table 3: Docker standalone cluster on one laptop (4 cores / 8 threads), all operations, 3 runs per size. Speedup and efficiency relative to 1 worker. For reference, `local[*]` takes 90.3 s.

6 Discussion

The optional operations. Both choices exposed the bottleneck they were picked for, though not always the expected one. O2's cost was recomputation and, once that was cached, task overhead: three outputs over one aggregate made it the only operation where caching paid off, and the cache's frozen 200 partitions then became its largest cost (Sections 4.2 and 4.4). O4's cost was the join back to every trip, removed by broadcasting the bucket table; the hot-zone skew it was chosen to probe did not appear at this scale (max/median task time ≤ 1.1).

Remaining bottlenecks. After the optimisations each operation is one scan plus small stages. The scan itself is now the cost, bounded locally by CPU decode and on the raw files by their 32 row groups; the per-action scheduling cost remains as a fixed part

of every run. O1 (outside the report scope) still spills 1.4 GB in a window over all trips.

Limitations. All numbers come from one laptop (`local[*]` or Docker), so shuffles never cross a network: the measured shuffle cost is a lower bound, and the cluster-size results reflect shared hardware rather than added nodes. Two series were rerun because unrelated load slowed every operation; differences below ~2 s are noise. Without a cluster, the 2016–2020 data was not processed, so effects that appear only at that scale (spill, skew) may be under-represented.

With more time. Measure on a real multi-node cluster and on 2016–2020; sort or bucket the cleaned copy by `PULocationID`.

References

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Individual contributions

Single-author project: Gabriel Matias Falcão, 100 %.

Comments and suggestions

Many more optimisations are possible, and the assignment's goal is to explore them. Unfortunately, both to time and report length constrains, many were left unexplored, nonetheless the optimisations implemented are modular through feature flags, some other's were implemented though not reported on, though still fun to explore and learn from. Whilst it is known that the DI cluster's unavailability is not the professor's fault, it is a pity that the assignment's main goal could not be fully explored.

Acronyms

AQE Adaptive Query Execution 1–3

BHJ Broadcast Hash Join 2

JVM Java Virtual Machine 3

SMJ Sort-Merge Join 1–3

TLC NYC Taxi and Limousine Commission 1